

Assignment 4: Sequence-to-Sequence Learning with LSTM

CS564

Deadline: 16 Nov

1 Objective

The goal of this assignment is to implement a **Sequence-to-Sequence (Seq2Seq)** model using **LSTM** networks for a simple **machine translation task**. You are expected to:

- Understand the encoder–decoder architecture using LSTMs.
- Implement a Seq2Seq model from scratch using Keras/TensorFlow.
- Train and evaluate the model on a small English–French sentence dataset.

2 Dataset

Use the **English–French sentence pairs** dataset available at: <https://www.manythings.org/anki/fra-eng.zip>

Details

- Use the file `fra-eng.zip` (contains `fra.txt`).
- Each line contains an English sentence and its French translation separated by a tab.
- Limit to the last 10,000 sentence pairs for faster training.

3 Tasks

Part A: Data Preparation

- Tokenize and clean both English and French sentences.
- Build vocabulary for both languages and convert words to integer sequences.
- Pad sequences to a fixed maximum length.
- Split data into 80% training and 20% testing.

Part B: Seq2Seq Model Implementation

- Implement an **Encoder–Decoder** model using:
 - `Embedding` and `LSTM` layers for the encoder.
 - `LSTM` and `Dense(softmax)` layers for the decoder.

Part C: Evaluation

- Evaluate the model on the test set by generating translations.
- Compute:
 - BLEU score (using `nltk.translate.bleu_score`)
 - Example input–output pairs for qualitative analysis.

Part D: Experiments and Discussion

- Vary the number of LSTM units ($\{128, 256, 512\}$) and observe changes in translation quality.
- Discuss how sequence length affect performance.

4 Experimental Settings

- Programming Language: **Python**
- Libraries: **TensorFlow / Keras, NumPy, Matplotlib, NLTK**
- Use a fixed random seed for reproducibility.

5 Deliverables

Submit a Jupyter Notebook (`.ipynb`) containing:

- Data preprocessing, model implementation, and training.
- Evaluation metrics and BLEU score.
- All outputs must be visible (do not clear cell outputs).
- Submission Link: <https://forms.gle/wCQJ2rnaJHRKggyA8>